Verdant Coil – Phase 1.4 Builder Mode Plan

This document outlines the detailed implementation plan for Phase 1.4 (Builder Mode – Local) of Verdant Coil. It is written for Godot 4.4.1 and assumes beginner-level development skills. The plan builds upon Phase 1.3's layered TileMap system and introduces local level construction with drag-and-place editing, biomass budgeting, validation, and save/publish functionality.

Stage 1 – TileSet Preparation

- Open the TileSet resource for Verdant Coil.
- Add Atlas from the roguelike spritesheet, set Tile Size = 32x32.
- Define Terrain Sets: Walls (single-terrain), Pools (two-terrain: Ground vs Acid, Ground vs Sticky).
- Tag all edge, corner, and center variants so Godot auto-connects shapes.
- Assign metadata to tiles per Phase 1.3 schema: walkable, digestible, hazard type, cost.

Stage 2 – Scene Setup

- Create BuilderMode.tscn.
- Root: Control (CanvasLayer for UI, Node2D for TileMap).
- TileMap layers: BASE (Flesh), WALLS (digestible/non-digestible), HAZARDS (acid, sticky), MARKERS (spawn, heartroot).
- Connect TileMap to prepared TileSet.

Stage 3 – Palette UI

- Add a PalettePanel.tscn following Verdant UI style guide.
- Add TextureButtons for: Flesh, Wall, Digest Wall, Acid Pool, Sticky Pool, Spawn, Heartroot.
- Highlight selected tool, update cursor mode accordingly.

Stage 4 – Editing Logic

- Convert mouse clicks to grid coordinates with local_to_map().
- Left click paints selected tile type, drag paints multiple.
- Right click erases tiles.
- Spawn and Heartroot limited to one each; overwrite existing if placed again.
- Walls and Pools use set_cells_terrain_connect() for auto-tiling.

Stage 5 – Biomass Budgeting

- Track biomass cost as sum of placed tiles using their metadata.
- Show counter in top-left UI (Biomass: current / cap).
- Prevent placement if over budget (flash counter red).

Stage 6 – Validation

- Add Validate button in UI.
- Checks: one spawn, one heartroot, biomass within limit, path exists between spawn and heartroot (flood fill over walkable cells).
- Show popup error messages if invalid.

Stage 7 – Save & Publish

- Save button exports JSON with grid data and metadata.
- Publish button copies JSON to Published/ directory (local stub).
- Ensure unique filenames for saves.

Stage 8 – Outcome Verification

- Paint walls and pools, confirm auto-tiling produces clean edges.
- Check biomass counter updates live.
- Place spawn and heartroot, validate ensures path exists.
- Save and reload JSON file, confirm map restores correctly.